

NETWORKS AND COMPLEXITY

Solution 22-8

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 22.8: Two nice results [Lvl. 4]

Show that the following holds for any reaction diffusion system:

- a) A steady state that is unstable in a single isolated node can never be stabilized by coupling multiple identical nodes together.

Solution

If the system is unstable in an isolated node, the node Jacobian $\mathbf{P}$ must have a non-negative eigenvalue. If we then consider a coupled system, every eigenvalue of $\mathbf{P} - \kappa\mathbf{C}$ is also an eigenvalue of the Jacobian $\mathbf{J}$. But because the Laplacian always has at least one eigenvalue $\kappa = 0$, every eigenvalue of $\mathbf{P}$ must also be an eigenvalue of $\mathbf{J}$. Since we know that $\mathbf{P}$ has a non-negative eigenvalue, also $\mathbf{J}$ must have a non-negative eigenvalue and hence the coupled system is unstable.

- b) Diffusion can never destabilize a homogeneous state if all variables diffuse independently and at the same rate, i.e. $\mathbf{C} = c\mathbf{I}$.

Solution

If we substitute $\mathbf{C} = c\mathbf{I}$ into $\mathbf{P} - \kappa\mathbf{C}$, it becomes $\mathbf{P} - \kappa c\mathbf{I}$, so we are subtracting a multiple of the identity matrix from the node Jacobian. But this is our eigenvalue shifting rule! The subtraction will simply shift every eigenvalue to the left by $c\kappa$. Therefore, if all variables diffuse at identical rates, increasing κ will uniformly decrease the real parts of all eigenvalues. Hence, no real part can increase as a result of an increase in κ , so diffusion can never destabilize the system.